



Abdul Rasak M R

Senior Software Engineer | Healthcare Integration & Microservices Specialist

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🎓 B.Tech CSE (MG University 2012)

Results-driven **Senior Software Engineer** with over **11+ years of experience** architecting and implementing enterprise Healthcare Information Systems (HIS/CMS), Pharmacy Management Solutions, and Supply Chain ERPs across UAE and India. Recognized specialist in regional healthcare interoperability (HAAD, DHA, Malafi, Nabidh, Riayati) using **HL7, FHIR messaging, and Mirth Connect**, alongside Laboratory Information System (LIS) device integrations. Currently spearheading end-to-end cloud-native microservices modernization using **Spring Boot, React.js, OAuth2, and PostgreSQL**.

11+

Years Software
Experience

5+

HIE Integrations (HAAD,
DHA, etc.)

4+

Enterprise Platforms
Architected

HL7 & FHIR

Interoperability Standards

🛠️ Work Experience

Senior Software Engineer

March 2015 – Present (11+ Yrs)

Enterprise Healthcare & IT Solutions

Career Progression: Junior Software Engineer (2.1 yrs) → Software Engineer (1 yr) → Senior Software Engineer (2018 – Present)

- ▶ **Legacy Modernization & Microservices:** Architecting and migrating monolithic enterprise healthcare applications to modern cloud-native **Spring Boot Microservices** with **React.js**

frontends, **OAuth2** security, and **PostgreSQL** databases.

- ▶ **Healthcare Interoperability:** Engineered bi-directional integrations connecting Cortex HIS/CMS with UAE regional Health Information Exchanges (HIE) including **Malafi** (Abu Dhabi), **Nabidh** (Dubai), and **Riayati** (MOHAP) leveraging **HL7 v2**, **FHIR**, and **Mirth Connect** engine.
- ▶ **Laboratory Equipment Integration:** Designed and implemented automated LIS device connectivity interfacing clinical laboratory machine analyzers directly with hospital systems using HL7 protocols.
- ▶ **Healthcare Claims & RCM Automation:** Developed automated insurance claims management workflows integrated with **HAAD** and **DHA** regulations, facilitating e-Claim Prior Approvals, Initial Submissions, Remittance Advice parsing, and Resubmissions.
- ▶ **iERP & Mobile API Development:** Created specialized supply chain and procurement management suite (iERP), integrating RESTful APIs to track and engage field worker activities via mobile apps.
- ▶ **Reporting Engine:** Designed extensive clinical, financial, and inventory analytical dashboards utilizing **Jasper Reports**.

Spring Boot

Microservices

React.js

OAuth2

Java 8 / Java 7

HL7 & FHIR

Mirth Connect

Spring MVC

Spring Security

MySQL & PostgreSQL

Jasper Reports

jQuery / JS

Freelance Web Designer & Developer

Jan 2013 – Feb 2015

Independent Contractor

- ▶ Designed and built custom, responsive client websites and web application interfaces from scratch using HTML5, CSS3, and JavaScript.
- ▶ Collaborated directly with clients for requirement analysis, UI wireframing, deployment, and performance optimization.

HTML5

CSS3

JavaScript

UI/UX Design

Responsive Web

Java Software Developer Intern

Sep 2012 – Jan 2013

Brazier IT Solutions, Bangalore

- ▶ Acquired practical hands-on experience in core Java application logic, Object-Oriented Programming (OOP) design patterns, and database manipulation.
- ▶ Assisted in module testing, bug fixing, and database query writing.

Java

OOP

SQL

Software Lifecycle

Major Projects Architected

Microservices Migration & LIS Integration Platform

Spearheading the architectural overhaul of enterprise suites into Spring Boot Microservices and React.js. Integrated Laboratory Information System (LIS) directly with medical auto-analyzers using HL7.

React.js

Spring Boot

Microservices

OAuth2

PostgreSQL

HL7

Cortex HIS & CMS (Hospital Information System)

Enterprise hospital and clinical management application. Integrated UAE Health Information Exchanges (Malafi, Nabidh, Riayati) and regional health authorities (HAAD, DHA) using HL7, FHIR, and Mirth Connect.

Java 8

Spring MVC

HL7 v2 & FHIR

Mirth Connect

MySQL

Jasper Reports

Pharmacy Management Application (UAE & India)

Integrated pharmacy operation platform handling inventory, drug dispensing, and automated insurance billing. Integrated HAAD & MOH prior approvals, claim submissions, and remittance collection resubmissions.

Java 7

Spring MVC

Spring Security

MySQL 5.6

Jasper Reports

jQuery

iERP - Procurement & Warehouse Management Suite

Dedicated supply chain and procurement system managing inventory workflows. Features RESTful APIs connecting mobile applications for real-time warehouse staff tracking and activity management.

Java 8

Spring MVC

MySQL

Jasper Reports

REST APIs

Mobile Sync

Technical Expertise

BACKEND & MICROSERVICES

Spring Boot

Microservices

Java 8 / Java 7

OAuth2

Spring MVC

Spring Security

RESTful APIs

HEALTHCARE INTEROPERABILITY

HL7 v2 Standard

FHIR Messaging

Mirth Connect

HAAD (Abu Dhabi)

DHA (Dubai)

Malafi HIE

Nabidh HIE

Riayati HIE

LIS Machine Integration

FRONTEND & WEB

React.js

JavaScript (ES6+)

jQuery

HTML5

CSS3

Responsive Design

DATABASES & REPORTING

MySQL (5.6+)

PostgreSQL

Jasper Reports

SQL Optimization

Education

B.Tech in Computer Science & Engineering

MG University, Kerala, India

Passout Year: 2012

Domain Specializations

- ✓ Hospital Information Systems (HIS)
- ✓ Clinical Management Systems (CMS)
- ✓ Healthcare Revenue Cycle (RCM)
- ✓ Laboratory Information System (LIS)
- ✓ Supply Chain & Warehouse ERP